



CATHOLIC JUNIOR COLLEGE
JC2 PRELIM EXAMINATION
Higher 1

**CANDIDATE
NAME**

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CLASS

2T

**INDEX
NUMBER**

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BIOLOGY

Paper 2 STRUCTURED & FREE RESPONSE QUESTIONS

8876/02

20 AUGUST 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [8]	
2 [12]	
3 [8]	
4 [5]	
5 [12]	
6 or 7 [15]	
TOTAL P2	60

This document consists of **16** printed pages and **0** blank page.

[Turn over

Section AAnswer **all** questions in this section

- 1 (a) Fig. 1.1 is an electron micrograph showing part of an organelle present in a mature plant cell.

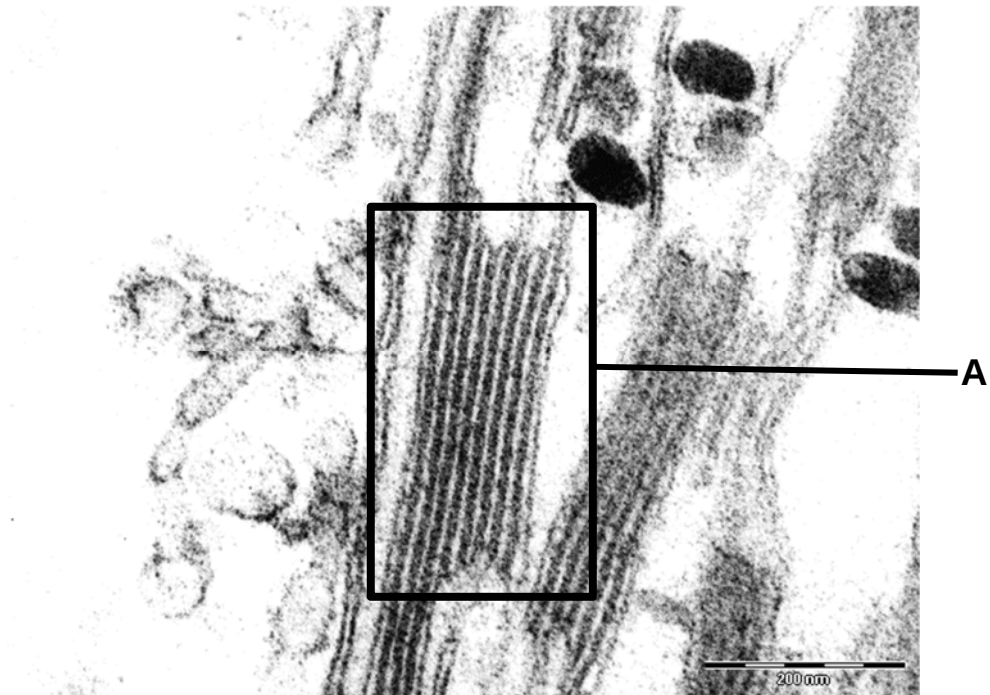


Fig. 1.1

- (i) Identify the organelle in which structure **A** resides.

.....[1]

- (ii) Explain the significance of the flattened stack arrangement in structure **A**.

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[2]

(b) Fig. 1.2 shows the structure of a cell surface membrane.

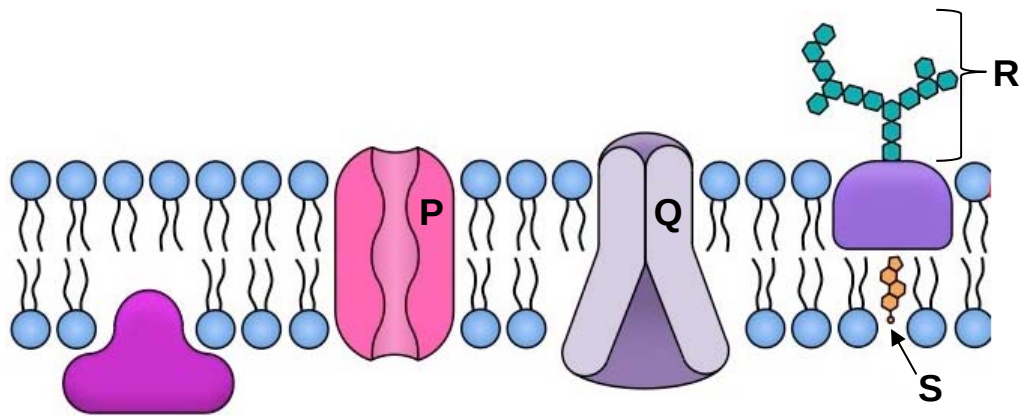


Fig. 1.2

With reference to Fig. 1.2,

(i) explain how the structure of **Q** allows it to be embedded in the cell surface membrane.

.....
[1]

(ii) explain how structure **P** functions differently compared to structure **Q**.

.....
[1]

(iii) explain why structure **S** is amphipathic.

.....
[1]

An immune response is triggered when the body recognises and defends itself against any substances that appear foreign to the body.

(iv) Suggest why a change in **R** may cause an immunological response.

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[2]

[Total: 8 marks]

- 2 (a) Fig.2.1 shows the structure of an adult haemoglobin molecule, HbA.

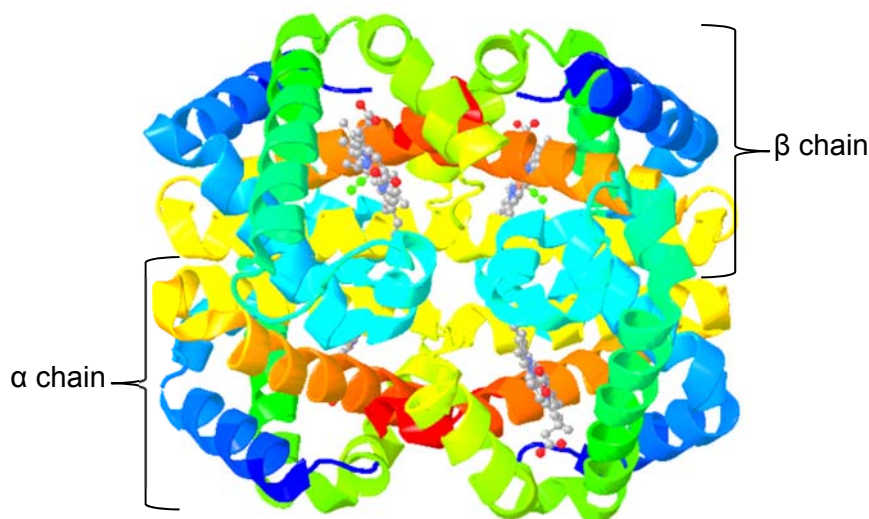


Fig. 2.1

- (i) With reference to Fig. 2.1, explain how the structure of HbA is related to its function.

.....

[2]

Sickle cell anaemia is an inherited disease due to a mutation in the allele that codes for β -globin polypeptide chain. This mutant HbS allele results in the production of haemoglobin variant HbS. Figure 2.2 shows the blast sequence of both the normal adult haemoglobin variant, HbA, as well as the mutant adult haemoglobin variant, HbS.

Beta globin sequence in normal adult haemoglobin (HbA)

nucleotide base	CTG ACT CCT GAG GAG AAG TCT
amino acid	Leu – Thr – Pro – Glu – Glu – Lys – Ser

Beta globin sequence in mutant adult haemoglobin (HbS)

nucleotide base	CTG ACT CCT GTG GAG AAG TCT
amino acid	Leu – Thr – Pro – Val – Glu – Lys – Ser

Fig. 2.2

- (ii) With reference to Fig. 2.1 and Fig. 2.2, explain how this mutation affects the structure of the haemoglobin molecule.

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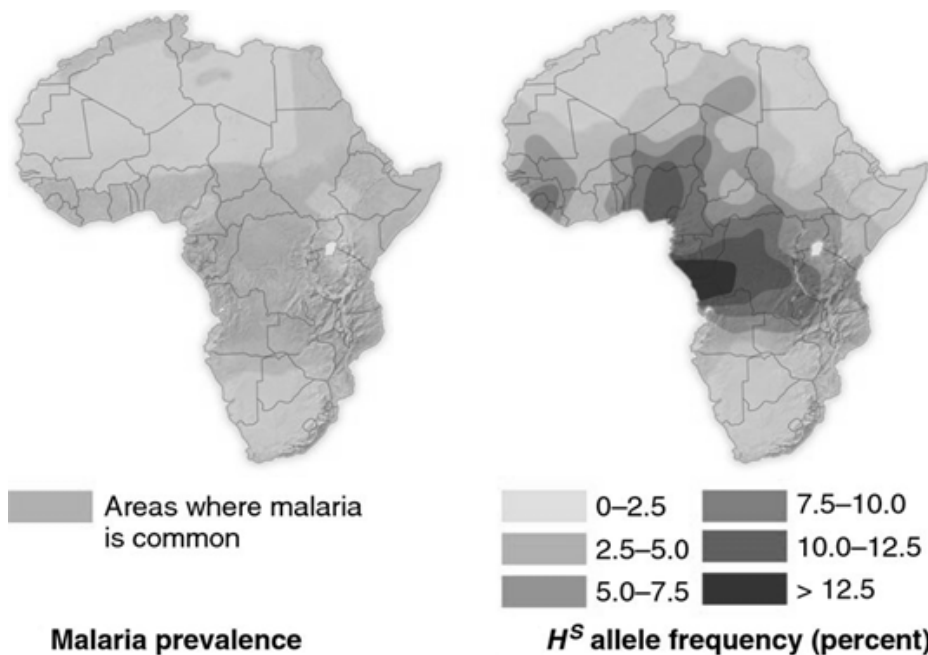
.....[2]

Sickle cell disease has its highest occurrence in sub-Saharan Africa. Despite the survival disadvantage of the allele HbS, HbS allele persists at more than 12.5 % in the population.

Fig. 2.3 shows regions where sickle cell anaemia is most common, which coincide with regions where malaria is. Malaria-causing parasites are introduced into the blood by mosquitos.

Fig. 2.3

- (iii) With
Fig.
why
still



reference to
2.3, explain
HbS allele is
preserved
within the
population and
suggest why it
occurs at an
unusually high

frequency in Africa.

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.....[4]

- (b) Haematopoietic stem cell (HSC), usually derived from bone marrow, peripheral blood or umbilical cord blood have shown great promise in the treatment of sickle cell disease recently.

Sources of such stem cells may be autologous (i.e. using the patient's own stem cells), allogeneic (i.e. using stem cells derived from a donor) or syngeneic (i.e. using stem cells from an identical twin).

After extraction, researchers cultivate these stem cells before injecting them back into the patient. Such haematopoietic stem cell transplantation (HSCT) therapy could potentially give new hope to patients with sickle cell anaemia, although the treatment comes with its own risk towards the patients.

- (i) Explain how extracted haematopoietic stem cells are cultivated before they are injected back to the patient.

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.....[2]

- (ii) Explain why treatment using HSC from allogeneic sources may pose some risks to the patient.

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.....[2]

[Total: 12]

- 3 Fig. 3.1 shows chickens with two different feather colours in which the gene for feather colour is carried on an autosome. The gene has two alleles, one that codes for black and the other for splashed-white. When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

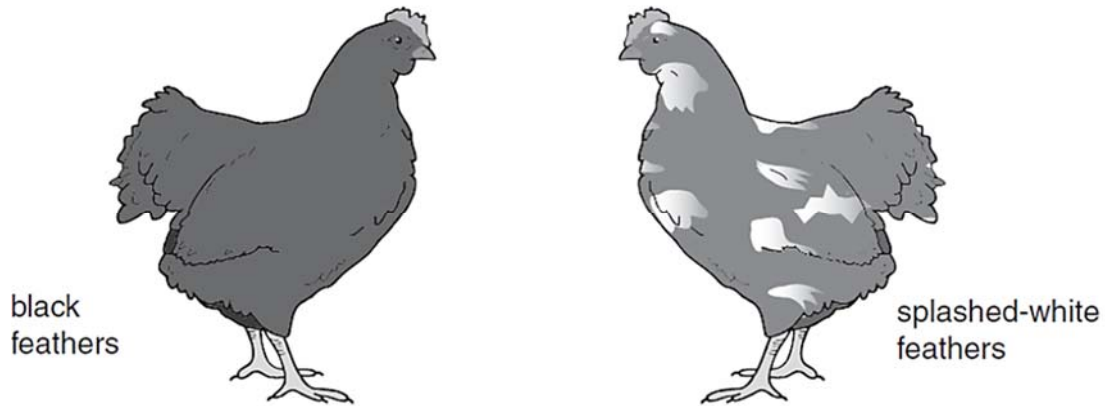


Fig. 3.1

- (a) The blue feathers is the result of codominance.

Explain what is meant by 'codominance'.

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.....[2]

Another gene may cause stripes on feathers (barred feathers), as shown in Fig. 3.2. This gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant over the allele for non-barred feathers (X^a).

In chickens, the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

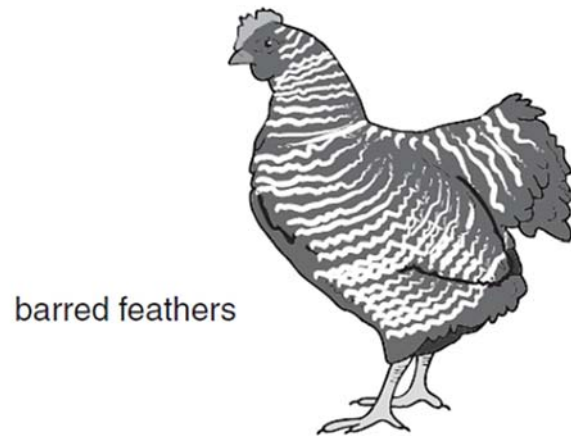


Fig. 3.2

- (b)** A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above draw a genetic diagram to show this cross.

- (c) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers.

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.....[3]

[Total: 8]

- 4 A student investigated respiration in a population of yeast growing in a sealed container.

Fig. 4.1 shows the results of his investigation.

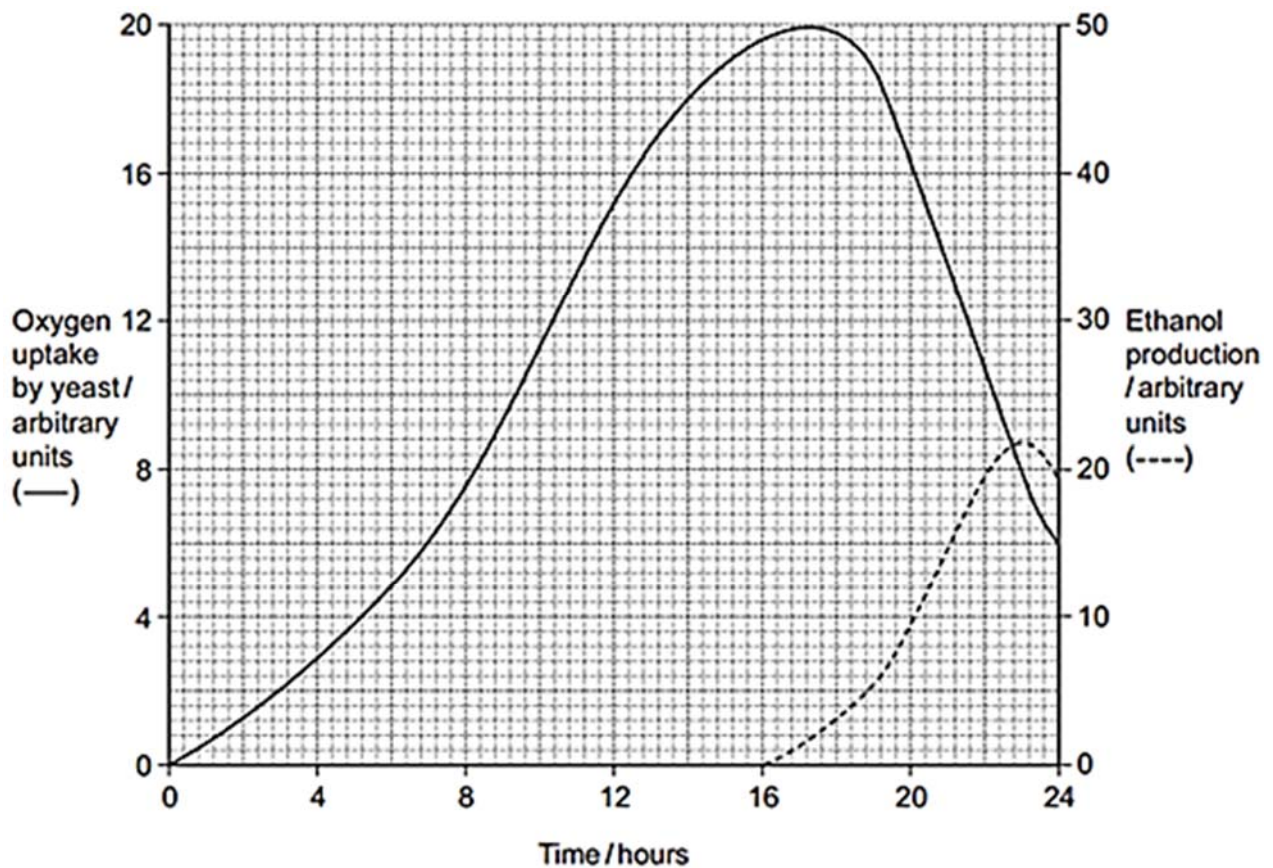


Fig. 4.1

- (a) With reference to Fig. 4.1, explain the changes in the ethanol production by yeast during this investigation.

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.....[3]

- (b)** Sodium azide is a substance that inhibits the electron transport chain in respiration. The student repeated the investigation but added sodium azide after 4 hours.

Suggest how the addition of sodium azide would affect oxygen uptake and ethanol production by yeast.

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.....[2]

[Total: 5]

- 5 Monarch butterflies are one of the largest migrating butterflies in the world. They have recently come under the spotlight in research due to how global warming is affecting their populations.

Monarch butterflies, shown in Fig. 5.1, are obligate feeders on a specific species of milkweed plant for both adults and caterpillars, and lay their eggs on these plants as well. These plants also secrete a toxin called cardenolides that deter most vertebrate herbivores that the monarchs are tolerant to. The monarchs even accumulate these toxins in their bodies that make them taste bad for their predators.



Fig. 5.1

- (a) Based on the information above, identify two selection pressures for the monarch butterflies.

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.....[2]

- (b) Explain how monarch butterflies and caterpillars have evolved to have a defense against their predators.

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.....[3]

- (c) Monarch butterflies, like most insects, are highly sensitive to weather and climate. Based on Q_{10} theory, they depend on environmental cues, temperature in particular, to trigger reproduction, migration, and hibernation.
- (i) State the meaning of Q_{10} theory and describe the correlation of this theory on insect physiology.

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.....[2]

Fig 5.2 below shows data of monarch butterflies arriving at Mexico during winter.

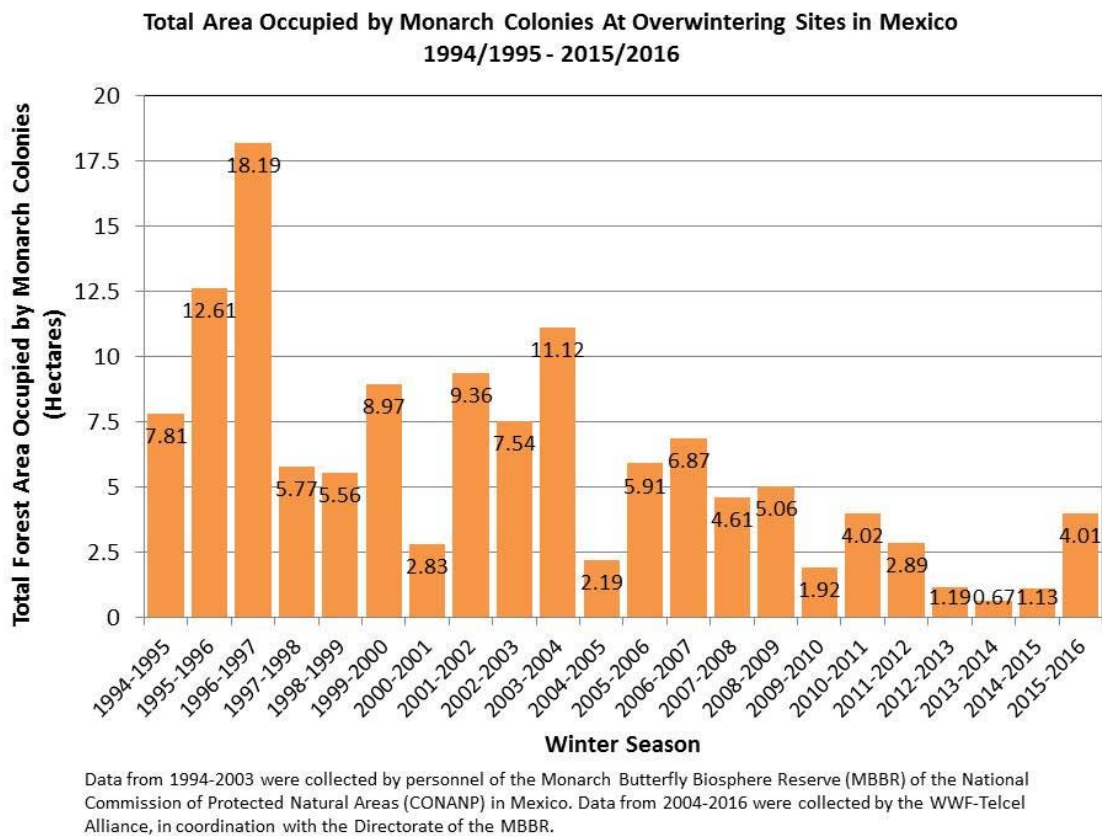


Fig. 5.2

- (ii) Explain why monarch butterflies would be highly sensitive to climate change.

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.....[2]

- (iii) Based on recent research, it has been found that increasing temperature can cause milkweeds to secrete a significantly higher level of cardenolides than normal.

With reference to Fig. 5.2 and this information, suggest how climate change is likely going to affect the monarch butterflies.

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.....[3]

[Total: 12]

Section B
Answer EITHER 6 OR 7.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 6** In photosynthesis, glyceraldehyde 3-phosphate (3-carbon sugar) is produced in the stroma of chloroplast during Calvin cycle.

(a) Outline this process and how the glyceraldehyde 3-phosphate is converted into starch and discuss the suitability of starch as a storage compound in plants.
[8]

(b) Compare the processes of Calvin cycle to that of the Krebs cycle. [7]

[Total: 15]

- 7 (a)** Outline the process of DNA replication and discuss the suitability of DNA as a hereditary material. [8]

(b) Compare the processes of DNA transcription to that of DNA translation. [7]

[Total: 15]

END OF PAPER